

FraudShield AI – Project Overview

Overview

FraudShield AI is an explainable fraud detection platform that analyzes suspicious digital content such as URLs, messages, and documents. It helps users assess whether a link, text, or document is risky before they act on it.

Project Objective

The main goal of the project is to build a local AI-powered security system that can detect phishing URLs, scam messages, and suspicious documents while remaining explainable to end users.

Core Features

- URL and link analysis for phishing detection
- Message and SMS scam detection
- Document risk assessment for PDF, PNG, and JPG files
- Explainable risk scoring
- User dashboard and admin analytics
- Downloadable PDF and HTML reports
- JWT-based authentication and secure access control

Tech Stack

The project combines React, TypeScript, FastAPI, Python, SQLAlchemy, PostgreSQL/SQLite, scikit-learn, Docker, and several document-processing libraries to build a complete AI-powered web application.

System Architecture

The application follows a layered architecture with a frontend, backend API, services, ML layer, and database layer. The frontend communicates with the backend through Axios and JWT-protected API calls.

How It Works

The user signs in, chooses the scan type, sends the input to the API, validates the payload, runs the feature extraction and ML models, computes the final risk score, and shows the results in a dashboard.

AI / ML Components

URL detection uses Random Forest over structural features; message detection uses TF-IDF plus Logistic Regression; document analysis checks metadata, structural anomalies, and OCR output when available.

Explainable Risk Engine

The engine creates a risk score, band, key indicators, plain-language explanation, and recommendations so the system is understandable rather than opaque.

Deployment and Testing

The system supports Docker deployment, PostgreSQL and SQLite fallback, and automated testing using pytest and smoke tests for live API behavior.

Conclusion

FraudShield AI is a full-stack, AI-powered fraud detection platform that combines machine learning, explainability, web development, backend engineering, and security into one functional project.